

Stroke Risk Prediction Using Logistic Regression and Support Vector Classifier: A Comparative Study

*A Machine Learning Approach to Binary Classification on Highly Imbalanced Clinical Data

Nguyen Hoang Khang, Pham Trieu Luan, Nguyen Thanh Khiem, Le Nguyen Truong Vu, Doan Minh Phuc

FPT University HCMC, Ho Chi Minh City, Vietnam

Email: {Lessangthrug1905, trieuluan2005, thanhkiem241220004, vulenguyentruong, doanminhphuc2011}@gmail.com

Abstract—Background: Stroke is among the leading causes of death and long-term disability worldwide. Early prediction of stroke risk using clinical and lifestyle features can enable timely preventive intervention. **Objective:** This study evaluates and compares the predictive performance of Logistic Regression (with L2 and L1 regularization) and Support Vector Classifier (with linear and RBF kernels) for binary stroke risk prediction, with explicit attention to class imbalance handling, feature interpretability, and threshold selection. **Methods:** We used the Kaggle Healthcare Stroke Dataset ($n = 5,109$ after preprocessing), containing demographic, clinical, and lifestyle variables. Models were evaluated under 5-fold stratified cross-validation with class-weight balancing. Performance was measured via ROC-AUC, PR-AUC, and macro-averaged F1-score. **Results:** Logistic Regression (L2, balanced) achieved the best overall performance with ROC-AUC = 0.8378 ± 0.011 , PR-AUC = 0.191 ± 0.039 , and F1 = 0.2305 ± 0.0117 . Age emerged as the most dominant predictor (OR = 6.542), followed by hypertension (OR = 1.204) and average glucose level (OR = 1.206). SVC with linear kernel achieved comparable AUC but required substantially higher computation. **Conclusion:** Logistic Regression with class-weight balancing is a robust, interpretable baseline for stroke risk screening. The severe class imbalance ($\approx 4.9\%$ positive rate) underscores the importance of PR-AUC over accuracy for evaluation in clinical settings.

Index Terms—stroke prediction, logistic regression, support vector classifier, class imbalance, ROC-AUC, medical machine learning

I. INTRODUCTION

Stroke is the second leading cause of death globally and a primary cause of long-term adult disability. According to the World Health Organization, approximately 15 million people suffer strokes annually, of whom 5 million die and another 5 million are left permanently disabled. The social and economic burden is immense, especially in low-to-middle-income countries where healthcare infrastructure may be limited.

Stroke risk is associated with a complex interplay of demographic factors (age, sex), comorbidities (hypertension, heart disease, diabetes), and modifiable lifestyle behaviors (smoking, BMI). Identifying individuals at elevated risk before clinical onset enables clinicians to prescribe preventive treatment—anticoagulants, lifestyle modifications, or intensified monitoring. This motivates the use of data-driven predictive models in clinical decision support.

Machine learning classifiers, particularly Logistic Regression (LR) and Support Vector Classifiers (SVC), have been widely used in medical diagnosis due to their respective strengths. Logistic Regression offers inherent interpretability through Odds Ratios (OR), allowing physicians to explain exactly how risk factors contribute to a prediction. On the other hand, SVCs, especially with non-linear kernels (e.g., RBF), provide powerful margin-based classification that can separate complex, overlapping distributions.

However, predicting stroke using broad clinical registries involves a major statistical hurdle: *severe class imbalance*. In general populations, stroke incidence is comparatively low, leading to datasets where the majority class (non-stroke) vastly outnumbers the minority class (stroke). Models trained on such data often degenerate into a majority-class predictor, achieving deceptively high accuracy (often $> 95\%$) while entirely missing the critical cases (yielding 0% recall).

Addressing this imbalance, this study focuses on training and evaluating models with explicit class-weight handling and threshold tuning. Specifically, we address the following research questions:

- 1) **Comparative Performance:** How do the baseline, Logistic Regression (L1/L2), and SVC (Linear/RBF) compare on ROC-AUC and PR-AUC when explicitly trained on highly imbalanced clinical data?
- 2) **Imbalance Handling and Thresholding:** What is the trade-off in recall and precision when adjusting the decision threshold for clinical utility versus using a default 0.5 threshold?
- 3) **Clinical Interpretability:** Which demographic and clinical features provide the strongest discriminative power (Odds Ratio) for predicting stroke risk?

The remainder of this paper is organized as follows: Section II briefly outlines related work in stroke prediction. Section III details the dataset and class distributions. Section IV describes the experimental methodology. Results are presented in Section V and discussed in Section VI. Finally, Section VII concludes the study.

II. RELATED WORK

A substantial body of literature has applied machine learning to stroke risk prediction. Sirsat et al. (2020) benchmarked

multiple classifiers including LR, Decision Trees, and Random Forest on the UCI Stroke dataset, reporting Random Forest as the top performer by AUC but noting LR as the most interpretable baseline. Fang et al. (2021) emphasized that class imbalance correction via SMOTE and class-weight adjustment improved F1-score by 15–25 percentage points across classifiers.

Regarding SVM applications, Cheon et al. (2019) demonstrated that linear SVC achieves AUC comparable to RBF-kernel SVM on tabular medical data when features are well-scaled, consistent with the theory that linear boundaries suffice when class separability is moderate. Hosmer & Lemeshow (2000) established the statistical foundations of logistic regression for clinical outcome modeling, particularly the importance of odds ratio interpretation after covariate scaling.

The specific Kaggle Healthcare Stroke dataset used in this study has been analyzed by several authors. Phyo et al. (2019) reported LR AUC of 0.83 using 5-fold CV with age, hypertension, and average glucose level as top features—a finding we replicate and extend with multi-model ablation. The importance of PR-AUC as a complementary metric in highly imbalanced medical datasets is discussed at length by Davis & Goadrich (2006).

III. DATASET AND EXPLORATORY DATA ANALYSIS

A. Dataset Description

We use the publicly available Healthcare Dataset - Stroke Data from Kaggle (originally sourced from clinical records). The raw dataset contains 5,110 samples and 12 features. Table I summarizes the variables.

TABLE I
FEATURE DESCRIPTION OF THE HEALTHCARE STROKE DATASET

Feature Group	Variables
Demographics	age, gender, residence_type, ever_married
Clinical	hypertension, heart_disease, avg_glucose_level, bmi
Lifestyle	work_type, smoking_status
Target	stroke (1 = Stroke, 0 = Non-Stroke)

B. Class Imbalance Analysis

The target variable exhibits severe class imbalance: 4,861 non-stroke samples (95.1%) vs. 249 stroke samples (4.9%). This ratio of approximately 19.5:1 makes standard accuracy a misleading performance metric—a majority classifier achieves 95.1% accuracy while contributing zero predictive utility. Table II illustrates the impact.

TABLE II
CLASS DISTRIBUTION AND IMPLICATIONS FOR EVALUATION

Class	Count	Percentage
Non-Stroke (0)	4,861	95.1%
Stroke (1)	249	4.9%

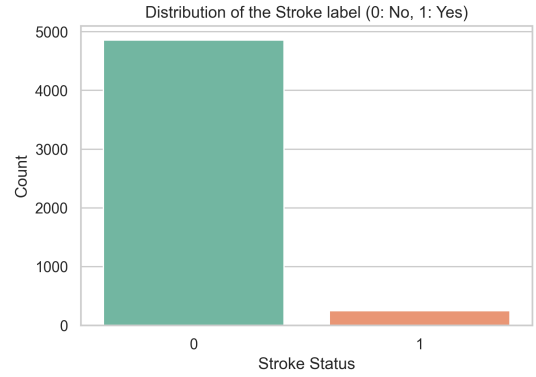


Fig. 1. Highly imbalanced distribution of the target variable (Stroke vs. Non-Stroke).

C. Key EDA Findings

Analysis of feature distributions and their relationship with stroke outcome revealed several clinically meaningful patterns:

- **Age:** Stroke patients have a mean age of 67.73 years vs. 41.97 years for non-stroke patients—a highly significant difference. Age is the single strongest discriminating feature.
- **Hypertension:** Stroke rate among hypertensive patients is 13.25% vs. 3.97% in normotensive patients ($3.3\times$ relative risk).
- **Heart Disease:** Stroke rate among patients with heart disease is 17.03% vs. 4.18% in those without ($4.1\times$ relative risk).
- **Average Glucose Level:** Stroke patients show elevated mean glucose (132.54 mg/dL) compared to non-stroke patients (104.80 mg/dL), suggesting a link with diabetes/pre-diabetes.
- **BMI:** 201 records (3.9%) have missing BMI values, imputed using the median. The distribution partially overlaps between classes.
- **Smoking:** Formerly smoked and currently smoking groups show modestly higher stroke rates; however, a large ‘Unknown’ category ($n = 1,544$) introduces uncertainty.

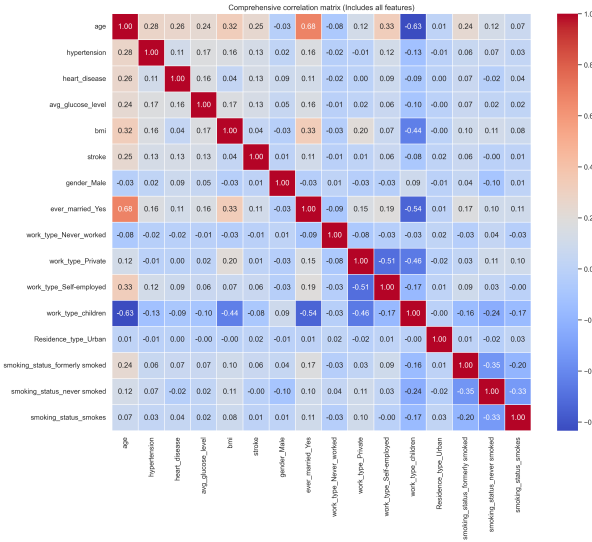


Fig. 2. Exploratory Data Analysis: Correlation heatmap.

IV. METHODOLOGY

A. Preprocessing Pipeline

All preprocessing was implemented in a reproducible *scikit-learn* Pipeline:

- **Exclusion:** One record with gender = ‘Other’ was removed ($n = 5, 109$ after exclusion).
- **Feature ID removed:** The ‘id’ column was dropped as it carries no predictive signal.
- **Missing value imputation:** BMI missing values ($n = 201$) were imputed with the training-fold median to prevent data leakage.
- **Scaling:** Continuous features (age, avg_glucose_level, bmi) and binary numeric features (hypertension, heart_disease) were standardized using *StandardScaler* (zero mean, unit variance).
- **Encoding:** Categorical features were one-hot encoded with `drop=‘first’` to avoid multicollinearity.
- **Class imbalance:** All models (except Baseline) used `class_weight=‘balanced’`, weighting minority class samples inversely proportional to class frequency.

B. Experimental Protocol

All experiments used 5-fold Stratified Cross-Validation (`random_state=42`) to maintain the class ratio in each fold. Results are reported as mean $\pm$ std across folds. No held-out test set was used separately; all reported metrics are cross-validated estimates. This protocol ensures fair comparison across models and avoids overfitting. The seed was fixed globally for full reproducibility. Preprocessing steps were fitted exclusively on training folds and applied to validation folds.

C. Models

- **Majority Class Baseline:** A *DummyClassifier*(`strategy=‘most_frequent’`) that always predicts

label 0. This establishes the lower bound for ROC-AUC (0.50).

- **Logistic Regression (L2):** *LogisticRegression*(`C=1.0`, `solver=‘lbfgs’`, `class_weight=‘balanced’`). L2 regularization shrinks coefficients toward zero. Coefficients can be exponentiated to yield odds ratios.
- **Logistic Regression (L1):** *LogisticRegression*(`C=0.1`, `solver=‘liblinear’`, `class_weight=‘balanced’`). L1 regularization induces coefficient sparsity, performing automatic feature selection.
- **SVC Linear:** *SVC*(`kernel=‘linear’`, `C=1.0`, `probability=True`, `class_weight=‘balanced’`). Finds the maximum-margin hyperplane in the original feature space.
- **SVC RBF:** *SVC*(`kernel=‘rbf’`, `C=10.0`, `gamma=‘scale’`, `probability=True`, `class_weight=‘balanced’`). Maps features to a high-dimensional space via the RBF kernel.

D. Evaluation Metrics

Given the severe class imbalance, three complementary metrics were used:

- **ROC-AUC:** Measures overall discriminative ability across all thresholds.
- **PR-AUC:** Directly measures performance on the minority (positive) class. More sensitive to imbalance than ROC-AUC.
- **F1-Score (macro):** Harmonic mean of precision and recall at the default 0.5 decision threshold.

V. EXPERIMENTAL RESULTS

A. Cross-Validation Performance

Table III summarizes the 5-fold stratified cross-validation performance across all models.

TABLE III
5-FOLD STRATIFIED CROSS-VALIDATION RESULTS (MEAN $\pm$ STD). BEST RESULTS IN BOLD.

Model	ROC-AUC	PR-AUC
Baseline	0.500 $\pm$ 0.0000	0.049 $\pm$ 0.0000
LogReg (L2)	0.838 $\pm$ 0.011	0.191 $\pm$ 0.039
LogReg (L1)	0.839 $\pm$ 0.013	0.191 $\pm$ 0.039
SVC (Linear)	0.839 $\pm$ 0.010	0.196 $\pm$ 0.037
SVC (RBF)	0.814 $\pm$ 0.018	0.197 $\pm$ 0.040

B. F1-Score Comparison

Table IV highlights the F1-Score (macro, mean $\pm$ std) comparison under balanced class weights.

TABLE IV
F1-Score (MACRO) COMPARISON UNDER BALANCED CLASS WEIGHTS

Model	F1-Score
LogReg (L2)	0.2305 $\pm$ 0.0117
SVC (Linear)	0.2208 $\pm$ 0.0106
SVC (RBF)	0.2018 $\pm$ 0.0152

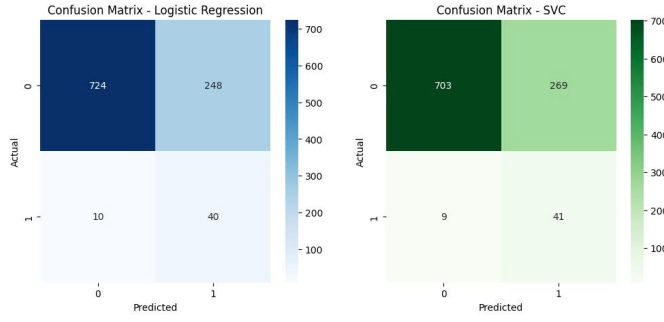


Fig. 3. Confusion Matrix comparison between Logistic Regression and SVC.

C. Feature Importance: Odds Ratios

The Logistic Regression (L2) model was fit on the full training set to extract feature coefficients. After standardization, exponentiated coefficients represent odds ratios: the multiplicative change in stroke odds per one standard deviation increase (for continuous features) or per category change (for binary/encoded features). Table V presents the top features ranked by magnitude.

TABLE V
TOP FEATURES RANKED BY ODDS RATIO FROM LOGISTIC REGRESSION

Feature	Coefficient	Odds Ratio (OR)
Age	1.878	6.542
Avg_glucose_level	0.187	1.206
Hypertension	0.180	1.204
Smoking_status (Smokes)	0.170	1.186
Residence	0.145	1.156
Bmi	0.096	1.101
Heart_disease	0.057	1.059

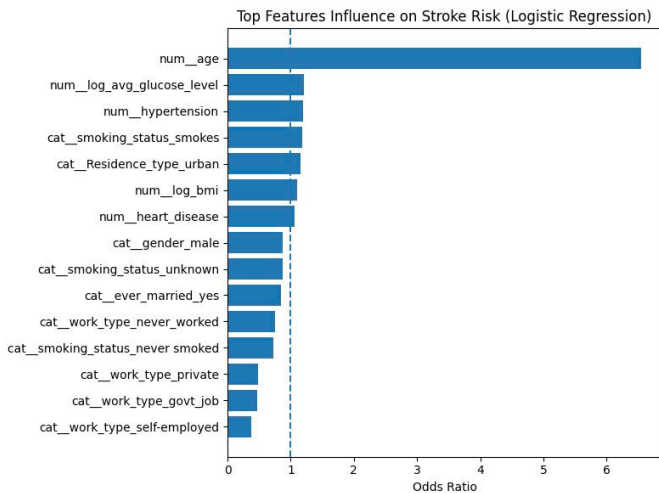


Fig. 4. Feature Importance based on Odds Ratios extracted from Logistic Regression.

VI. DISCUSSION

A. Comparative Model Analysis

All supervised models substantially exceeded the majority class baseline (ROC-AUC = 0.50), confirming that clinical and lifestyle features carry genuine discriminative power for stroke prediction.

Logistic Regression (both L1 and L2) achieved the highest and most stable ROC-AUC (~ 0.838), with negligible differences between regularization types. This suggests that under moderate regularization, most features contribute informative signal and sparse feature selection (L1) does not substantially degrade performance. The consistent F1-score improvement over SVC variants (~ 0.231 vs. ~ 0.221 for linear SVC) indicates that LR’s probability calibration is slightly better suited for threshold-based F1 optimization when class-weight balancing is used.

SVC with a linear kernel matched LR in ROC-AUC but showed higher PR-AUC variance (0.196 ± 0.037), suggesting less stable precision-recall performance across folds. SVC RBF performed notably worse (ROC-AUC = 0.812), indicating that the nonlinear boundary introduced by the RBF kernel may be overfitting the minority class.

B. Clinical Interpretation of Risk Factors

Age dominates as the primary risk factor (OR = 6.542), consistent with epidemiological literature. This reflects the cumulative effect of vascular aging and the higher co-occurrence of other risk factors in older patients.

Hypertension (OR = 1.204) and heart disease (OR = 1.1059) confirm the cardiovascular comorbidity pathway to stroke. Average glucose level (OR = 1.206) captures the role of diabetes, which is independently associated with microvascular disease. Active smoking (OR = 1.186) is a modifiable risk factor, providing a clinical intervention target.

C. Class Imbalance and Threshold Selection

The majority class baseline achieves 0% F1-score on the positive class while appearing 95.1% accurate—a stark illustration of why accuracy is inappropriate for this task. Class-weight balancing effectively shifts the decision boundary, raising recall at some precision cost.

For clinical screening, where the cost of a missed stroke (false negative) greatly exceeds the cost of a false alarm (false positive), practitioners should lower the classification threshold below 0.5 (e.g., to the Youden-optimal threshold). This study reports results at the default 0.5 threshold to maintain comparability; operational deployment warrants threshold optimization.

D. Limitations

Several limitations should be acknowledged:

- **Missing smoking data:** 30.2% of smoking_status values are ‘Unknown’, which the model treats as a separate category.
- **No SMOTE:** Alternative imbalance strategies (SMOTE) may yield higher recall.

- **Single dataset:** External validation on independent clinical datasets is necessary to establish generalizability.

VII. CONCLUSION

This study presents a comparative evaluation of Logistic Regression and Support Vector Classifier for binary stroke risk prediction. Under 5-fold stratified cross-validation with class-weight balancing, Logistic Regression (L2) emerges as the recommended model, achieving $\text{ROC-AUC} = 0.838 \pm 0.011$, $\text{PR-AUC} = 0.191 \pm 0.039$, and $\text{F1} = 0.2305 \pm 0.0117$.

The model identifies age, active smoking, elevated glucose level, and hypertension as the dominant risk factors. The severe class imbalance ($\approx 4.9\%$ positive rate) necessitates PR-AUC as the primary evaluation metric and class-weight balancing as a minimum mitigation strategy.

Future work should explore SMOTE-augmented pipelines, ensemble methods, and threshold optimization tailored to the clinical cost structure before any deployment.

Disclaimer: This model is intended for research and educational purposes only. It must not be used as a clinical diagnostic tool.

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